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Gen-AI's effects on new value propositions in business model innovation: Evidence from information technology industry

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ABSTRACT

Generative AI (Gen-AI) with its evolving natural language capabilities is dramatically changing the way that businesses operate and customers consume their products and services. While existing literature discusses Gen-AI's impact on computer science and engineering, its adoption significantly influences business models across various industries. This paper focuses on how Gen-AI affects new value propositions within business model innovation (BMI). The qualitative research method is adopted in this research. The data is collected and analyzed through 32 semi-structured interviews and archival sources. The study identifies five approaches — knowledge querying-based cloud solutions, content creation, AI agents, foundation models, and upstream industry chain infrastructure — that Gen-AI affects new value propositions in BMI. This research introduces empirical evidence from the information technology (IT) industry, broadening the contextual boundaries of Gen-AI's new value propositions in BMI. The study advances beyond isolated mechanisms, providing a quadrant view and process map to illustrate the interrelated dynamic effects of Gen-AI's new value propositions in both radical and incremental BMI.

1. Introduction

With the development of large language models (LLMs) such as ChatGPT[1] and Gemini[2] and content creation services such as Midjourney,[3] Pika[4] and Sora,[5] generative AI (Gen-AI) has achieved significant milestones. These LLMs and services showcase AI's evolving capabilities, opening new possibilities for natural language understanding and interactions.

The rapid advancement of Gen-AI is a focal point of current research, particularly within the realms of computer science and engineering (Wamba et al., 2023; Jo, 2023; Dwivedi et al., 2023). As Gen-AI continues to evolve, its transformative influence extends far beyond technical domains, resonating across diverse industries and businesses (Wessel et al., 2023; Chui et al., 2022). It could provide new possibilities for the existing business models in the form of business model innovations (BMI) (Haefner and Gassmann, 2023). This interdisciplinary impact underscores the pervasive nature of Gen-AI, shaping the technological landscape and fostering innovation and adaptation across a spectrum of industries.

Despite the recognition of Gen-AI's impact on business models in existing research (Haefner and Gassmann, 2023; Kanbach et al., 2023). Specifically, current studies acknowledge the influence of Gen-AI on business models in fields such as content creation (Epstein et al., 2023; Wessel et al., 2023) and education (Baabdullah, 2024; Michel-Villarreal et al., 2023). Furthermore, there are also future research opportunities for researching the BMI possibilities of Gen-AI based on the roadmap proposed by Mariani & Dwivedi (2024). There remains a gap in understanding the specific approaches and mechanisms through which Gen-AI shapes these changes, particularly from Gen-AI solution provider's perspective. As these domain-specific innovations from Gen-AI would not even be possible without the upstream AI solution provider, this research's motivation is to see the upstream innovations of Gen-AI, to better understand Gen-AI innovations in various domains. The theoretical perspective we adopt is to disentangle novel value propositions introduced by Gen-AI in the ever-changing landscape of

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BMI. It is because the value proposition is the first and foremost element of the BMI. This exploration would not only enhance the overall understanding of Gen-AI's influence but also reveal inventive applications and potential disruptions within diverse business landscapes, as various industries may encounter distinct challenges and opportunities in integrating Gen-AI, requiring a nuanced comprehension of its implications.

Therefore, the research objective is to disentangle new value propositions from Gen-AI to enable BMI to provide a perspective that better understands Gen-AI's potential in BMI. Following that, the research question is "How does Gen-AI affect new value propositions in BMI in the information technology (IT) industry, and how do they work?". By leveraging semi-structured interviews and archival sources, the study gathers and analyzes data to unearth valuable insights. Moreover, the research's contribution is to provide a relational map to understand what new value propositions can be achieved with Gen-AI in achieving incremental and radical BMI (Velu, 2015; Friedman et al., 2008) in the IT industry and how they work dynamically. This research's practical implication is to provide the framework for businesses to understand how Gen-AI enables new value propositions, offering an ecosystem view, mapping, and relational map to guide informed decision-making and drive BMI.

The paper is organized as follows: First, it provides the literature review, positioning Gen-AI in the digital business context, the value propositions in BMI, and Gen-AI's influence on BMI. Secondly, it justifies the methodological background, detailing the methodology rationale, data collection, and analysis processes. Furthermore, the study results are presented within the analysis framework, disentangling the proposed theoretical dimensions and the relational map. Finally, a comprehensive discussion is with respect to the contribution to literature, theoretical implications, and contribution to practice, followed by the conclusion.

2. Literature review

2.1. Gen-AI in digital business context

Gen-AI has rapidly emerged as a cornerstone technology, significantly influencing both digital business and IT industries (Susarla et al., 2023). This subsection incorporates various scholarly discussions, focusing on integrating Gen-AI in business operations, its impact on organizational structures, human resource management, and the pertinence of Gen-AI in evolving business models, especially regarding value propositions in BMI. These themes are determined based on substantive themes arising during the review of the literature following recommendations from Tranfield et al. (2003).

The remarkable growth of Gen-AI, exemplified by advancements like ChatGPT, has triggered significant debates and transformative trends in society and business. Studies by Wamba et al. (2023), Richey Jr et al. (2023), and others have highlighted the pivotal role of Gen-AI in revolutionizing operations and supply chain management. These works underscore the enhanced efficiency and performance improvements in supply chains attributed to Gen-AI adoption. They reveal a consensus on the benefits of Gen-AI, such as improved operational efficiency, but also note prevalent concerns around security, risks, and ethical issues. Despite these insights, there remains a notable lack of empirical evidence explicitly detailing Gen-AI's comprehensive effects, particularly in the operational and supply chain contexts.

In the realm of organizational adoption and integration of Gen-AI, works by Agrawal (2023) and AL-khatib (2023) stand out. They explore the factors impacting the acceptance and implementation of Gen-AI tools in organizations, using frameworks like the Technology-Organization-Environment framework. These studies contribute significantly to our understanding of organizational dynamics surrounding Gen-AI adoption, emphasizing the need for deep integration of Gen-AI to realize its full potential. However, they also identify a gap in understanding the nuanced implications of Gen-AI in different organizational contexts.

Gen-AI's impact on human resource management and strategic decision-making has been a focal point of research. Budhwar et al. (2023) and Norback & Persson (2023) delve into these aspects, discussing the potential for job displacement or creation and the shifts in labor dynamics. Their findings highlight the need for further more empirical research to understand Gen-AI's long-term impacts on the workforce and managerial practices. This area is particularly crucial as it directly influences how organizations structure their human resource policies and strategies in response to Gen-AI integration.

In recent academic discourse, there has been a nuanced examination of the complexities surrounding Gen-AI. A recent editorial from the Academy of Management Journal provides a discussion about the scenario-based illustrations of the impact of Gen-AI on the academic community (Grimes et al., 2023). Moreover, scholars have cautioned against the prevailing excitement about Gen-AI, advocating for a more balanced and grounded approach that acknowledges its potential and limitations (Vinsel, 2023). Ethical concerns, particularly in AI, have been revisited, suggesting that while the technology is new, its ethical dilemmas are not entirely unprecedented, thus necessitating a reevaluation of existing ethical frameworks (Niederman and Baker, 2023). The impact of Gen-AI tools on human cognition and scientific integrity has also been a point of contention, with concerns raised about the potential undermining of critical thinking and reflexive capabilities in research and management practices (Lindebaum and Fleming, 2023). Furthermore, the role of Gen-AI in academic research, specifically its function as a research assistant or co-author, has been critically analyzed, raising questions about the boundaries of AI-assisted content creation and its implications for authorship and intellectual contribution in scholarly work (Davison et al.). This body of literature collectively underscores the need for a comprehensive understanding of Gen-AI's multifaceted impact, encompassing ethical, cognitive, and authorship-related challenges in the academic realm.

In summary, while existing literature provides insights into Gen-AI's impact on content creation (Epstein et al., 2023; Kanbach et al., 2023; Wessel et al., 2023), higher education (Michel-Villarreal et al., 2023; Noy and Zhang, 2023), operations (Wamba et al., 2023), and IT (Ooi et al., 2023), there is a need for more empirical research, particularly case studies, to understand how Gen-AI transforms value propositions and enables BMI in these domains, aligning with the authors' objective to deeply explore Gen-AI's effect on new value propositions in BMI within the IT industry context.

2.2. Value propositions in business model innovation

Business models describe a company's architecture, logic, value proposition, and value creation and capture (Baden-Fuller and Morgan, 2010; Teece, 2010). Thus, BMI requires new value proposition, creation, and capture methods (Markides, 2006). BMI can redefine a product or service, how it is delivered, and how to monetize the customer value proposition. BMI can be incremental or radical (Velu, 2015). Specifically, incremental innovation refers to the implementation of small or minor modifications to an existing product or process in order to take advantage of the capabilities of an existing design (Friedman et al., 2008; Velu, 2015). Radical innovation involves making substantial deviations from the current design and has the potential to create new opportunities in terms of applications and markets (Friedman et al., 2008; Velu, 2015). Moreover, Velu (2018) proposed the 4V's of a business model, which are value proposition, value creation (Amit and Zott, 2001), value capture, and value network. A business model outlines the customer value proposition, the methods for value creation, the necessary network of partners, and the strategy to capture a portion of the created value for firms.

The value proposition is the focus of this research because it is the foundation for other "V"s in BMI. That is to say, without a clearly demonstrated value proposition, it is not possible to have a well-performed value creation, value capture, and value network. In the literature, there is a foundational theme that revolves around the centrality of value propositions in BMI (Menville and Kraemer, 2004; Kiel et al., 2017). Lehoux et al. (2014) assert the significance of value propositions in shaping both business model development and technological innovation, particularly in health technology. This perspective is furthered by Viswanadham (2018), who underscores the importance of customer-centric and value-driven business models. Additionally, Cheng and Wang (2022) explore how digital innovation attributes interplay with IT infrastructure to enhance BMI. It reveals that organization and product-oriented strategies should be focused on value propositions and user experience. Moreover, the case of Bosch further emphasizes the core value propositions in the business model change (Leiting et al., 2022).

The second theme highlights the adaptability of value propositions in response to technological and market changes. Nielsen and Lund (2018) discuss the scalability of business models, emphasizing the need for adaptable value propositions across different market segments. This idea resonates with Johnson, Christensen, and Kagermann (2008), who advocate for reevaluating business models in light of new technologies. In parallel, Winterhalter et al. (2017) contribute to this theme by examining business models for frugal innovation, indicating that innovative value propositions can create new markets, particularly in resource-constrained environments.

The third theme concerns the influence of digital transformation on value propositions within business models. Abbate et al. (2023) show how digital transformation steers the apparel industry towards sustainable business models, signifying evolving value propositions. Similarly, Chan, Bharadwaj, and Varadarajan (2023) provide empirical evidence on how digital innovation reshapes customer value. This theme is further enriched by Kazantsev et al. (2023), who emphasize the role of data in enhancing value propositions in business models, and by Brea (2023), who proposes a framework for mapping actor roles in digital ecosystems, highlighting how different roles can innovatively drive the ecosystem's value proposition.

However, recent research indicates that productivity gains from Gen-AI — such as higher-skilled worker output or improved creativity in customer-facing roles — can also be reframed as new ways to create and capture value. In other words, these "performance improvements" may lead to novel service offerings or revenue models rather than simply optimizing existing processes, thus bridging the gap between worker productivity and genuine BMI.

These themes collectively paint a comprehensive picture of the dynamic nature of value propositions in BMI. The literature suggests that businesses need to continuously innovate and adapt their value propositions to ensure alignment with evolving customer needs, technological capabilities, and societal shifts. However, there is little empirical evidence targeting the effects of Gen-AI on new value propositions in BMI in the IT industry.

2.3. Gen-AI's influence on BMI

There are three bodies of literature related to Gen-AI's influence on BMI, which are office work, education, and healthcare, illustrated as follows. The first body of literature is about office work, which includes the field experiment in the consulting (Dell'Acqua et al., 2023) and telemarketing field (Jia et al., 2024); the theoretical framework on the knowledge and creative work (Benbya et al., 2024); and the practical originated paper (Eapen et al., 2023).

Based on the field experiment with BCG consulting, the researchers argue that generative AI can boost highly skilled workers' performance by 40% (Dell'Acqua et al., 2023). While initially framed as a productivity improvement, these findings can also be interpreted as providing firms with new avenues for value creation and capture—such as AI-driven consulting services or strategic advisory offerings that build on augmented human expertise. Additionally, Jia et al. (2024) demonstrate that AI-generated sales leads elevate employees' creativity in customer interactions, which can be leveraged to develop innovative customer-facing solutions and differentiated value propositions (rather than simply improving efficiency). Theoretically, Benbya, Strich & Tamm (2024) recently provides a research agenda for Gen-AI and knowledge work, calling for attention to extensions on knowledge creation, retrieval, sharing, and application perspectives (Eapen et al., 2023). Practically, Eapen et al. (2023) argue that Gen-AI's potential lies in assisting humans in creating unimaginable solutions rather than replacing them.

Based on the field experiment with BCG consulting, Dell'Acqua et al. (2023) find that generative AI augments performance for highly skilled professionals. While initially framed as a productivity improvement, these findings can also be interpreted as providing firms with new avenues for value creation and capture—such as AI-driven consulting services or strategic advisory offerings that build on augmented human expertise.

The second group deals with the implications of Gen-AI in education. Studies in this category explore practical applications and potential misuses of Gen-AI in educational and managerial contexts (Baabdullah, 2024; Megahed et al., 2023; Michel-Villarreal et al., 2023). They provide important insights into the opportunities and ethical challenges Gen-AI poses, particularly in its application to educational strategies and management practices (Burger et al., 2023). This research is vital for understanding how educational institutions and management bodies should approach and integrate Gen-AI into their systems.

The third body of literature is related to the effects of Gen-AI on the healthcare field. For example, a key area of interest is using Generative Adversarial Networks (GANs) to model complex societal behaviors, such as those observed during public health crises (Bao et al., 2022). These papers highlight Gen-AI's ability to simulate human interactions and responses under various scenarios. Additionally, significant research on creating synthetic data while ensuring privacy is a critical concern in fields like healthcare (Gonzales et al., 2023). These papers reflect Gen-AI's capacity for handling sensitive information securely. Further, the range of Gen-AI applications extends to areas like language modeling and advanced visual simulations (Soliman and Al Balushi, 2023), illustrating the technology's diverse potential. These technical papers are foundational to enhancing our understanding of Gen-AI's capabilities and its role in driving innovative business models.

In conclusion, the literature on Gen-AI's impacts on BMI reflects diverse perspectives, from office work, education, and the healthcare field, providing the industry domain-specific arguments for demonstrating the potential of Gen-AI. Each domain illustrates unique pathways through which Gen-AI impacts business models, necessitating a deeper exploration of these avenues. Although it is exciting to explore the possibilities of domain-specific innovations, it is important to acknowledge an assumption that domain-specific innovations are dependent on the upstream solution providers in the context of Gen-AI. For example, without the advancement of the computing infrastructure, and the modern machine learning algorithm, Gen-AI capabilities (such as ChatGPT) in specific domains would not be possible. Therefore, our focus is on understanding the upstream innovations that drive downstream innovations across various domains. However, limited research focuses on the social-technical perspective of solution providers of Gen-AI applications or products (Stohr et al., 2024; Baabdullah, 2024; Akhtar et al., 2024; Fosso Wamba et al., 2023). The social-technical perspective is of vital importance, as it goes beyond Gen-AI as a technology, and embeds it as the products or services in delivering the focal companies' unique value propositions. That is to say, the IT companies that enable these Gen-AI services and their unique value propositions allow the achievement of BMI across industries. That is the rationale for choosing the IT industry as the focus of this research. That leads to the research gap due to a

lack of evidence on Gen-AI's effects on new value propositions in BMI with empirical evidence in the IT industry.

Based on the literature review, the research question is "What are the new value propositions enabled by Gen-AI to achieve BMI in the IT industry, and how do they dynamically interact?"

3. Research design

3.1. Rationale

This research applies the exploratory qualitative approach, as it allows for an understanding of complex issues that cannot be obtained through fixed-choice investigations (Miehé et al., 2023; Querci et al., 2024; Gopaldas, 2016). A qualitative investigation allows for the exploration of the complexities of Gen-AI and BMI to be revealed (Denzin and Lincoln, 2000). Respondents are encouraged to express their opinions, views, and experiences freely without any limitations. Moreover, qualitative research is particularly useful for studying complicated problems (Graebner et al., 2012; Marshall and Rossman, 2014) and for gaining deep insights into relatively new and under-researched areas (Eisenhardt and Graebner, 2007).

Specifically, this study uses case study research since it allows the researcher to capture real-world activities' extensive and essential characteristics (Abi Saad et al., 2024; Ghosh et al., 2022; Yin, 2013a). A case study comprises the following elements: (1) face-to-face interactions, such as interviews; (2) many individuals in a single organization; and (3) ongoing data collection that covers both current and past events (Easterby-Smith et al., 2015). To conduct the case study, the researchers carried out interviews in diverse industries, serving as divergent information sources to triangulate the findings, which enhances its generalizability (Eisenhardt and Graebner, 2007). This method offers the benefit of analyzing data within and across diverse scenarios and exploring multiple cases to identify similarities and differences, which is critical for exploring Gen-AI's effect on new value propositions in BMI.

At this stage, the advancement of Gen-AI is predominantly driven by developments within the upstream IT industry. As the underlying technologies continue to evolve, corresponding domain-specific innovations are expected to emerge. The rationale for focusing on the IT industry lies in its foundational role; a deeper understanding of upstream IT innovations provides valuable insights into the downstream, domain-specific applications that build upon them. Given the current phase of technological development, it is most appropriate to concentrate on the IT sector, as subsequent domain-specific advancements will likely draw heavily on these upstream innovations for their conceptual and practical frameworks.

3.2. Data collection

From March 2023 to December 2023, the research team conducted face-to-face interviews with teams from various companies, including business directors, business development managers, marketing managers, and strategic managers. No restrictions were imposed on the companies' geographic location, business scale, or ownership type (private vs. public). The team contacted 53 companies through email and attending academic conferences and industrial exhibitions, of which 32 (60%) agreed to participate in the interviews. These 32 companies included seven small-sized, eight medium-sized, and 17 large-sized companies. Among the selected samples, the theoretical saturation has been reached as there have been repeated patterns occurring (Glaser and Strauss, 2017; Gioia et al., 2013b). The selected and interviewed cases represent the companies in the IT industry that provide Gen-AI-related solutions to provide new value propositions. They are most likely to be SaaS companies, but they also include PaaS or computing hardware solution providers, etc. They are mainly in the UK, EU, and US because of two reasons. Firstly, these regions are where Gen-AI-related technologies emerged. Secondly, the diffusion of these innovations has been embedded in the way that these companies operate, which are more exposed to the researchers. We can foresee in the future, these innovations to be diffused in other places of the world. One to four experienced informants from each company participated in the interviews. While all of the companies adopted Gen-AI, most of them changed their value propositions in business models. Most of the non-participants rejected the interviews due to their lack of adoption of Gen-AI and new business models, their current busy work schedules, or strict confidentiality requirements. While these cases are representative of Gen-AI solution providers in the IT industry, it is important to acknowledge that further samples from more diverse geographical areas could enhance the generalizability of the findings of this research.

Data were collected from the 32 participating companies, and anonymity was offered to each company to ensure an open discussion of sensitive information. Additionally, the research team promised not to receive funding from any customer or governmental institution and that the information would not be shared with customers to ensure confidentiality.

Information was gathered from two distinct sources to ensure the validity and reliability of the qualitative research (Yin, 2013). The primary source of information was semi-structured interviews, while the secondary source was archival data obtained from various sources, including corporate reports, industry reports, public databases, newspapers, and other relevant materials (Seen Table 4 in Appendix D). The combination of these different data sources allowed for triangulation of information about Gen-AI and BMI within and across companies, thereby increasing the overall quality of the research. Specifically, we created a document for each interview and, when needed, requested additional information to validate our findings (Yin, 2013; Stohr et al., 2024). Moreover, data from the case studies were examined to develop explanations for each case, exploring and identifying the relationships between new value propositions in enabling BMI (Yin, 2013; Ozdemir et al., 2024).

3.2.1. Semi-structured interviews

A total of 32 interviews were carried out, with each company having at least one representative participating as informants. The interview protocol used is included in Appendix A. Most of the companies' headquarters were in the UK and US, while others were in Germany, France, Israel, the UAE, Japan, and South Korea. All the companies had business in the UK, and all interviewees were fluent in English. Some of the interviews were recorded with the interviewees' consent, and all interviews were documented and transcribed to facilitate the coding process.

3.2.2. Archival data

Various types of information were collected for this study, including internal sources such as consulting reports, performance evaluations, scorecards, monitoring dashboards, brochures, and others. External sources were also gathered, including corporate and industry reports, newspapers, and archival databases such as Factiva, Company House, Bloomberg, Economist, Wall Street Journals, Forbes, and LinkedIn. Some companies were publicly listed, and thus, much information was available from public sources. Occasionally, third-party reports from organizations like Statista were gathered to supplement the interview information. The archival data was useful in triangulating the information collected from the interviews and helped to focus the interviews on specific themes. Additionally, archival data helped extract valuable insights from representative companies that had not participated in the interviews.

The data source of this study with both primary interviews and archival data is shown in Table 4.

3.3. Data analysis

To answer the research questions (What are the new value propositions enabled by Gen-AI to achieve BMI in the IT industry, and how do they dynamically interact?), the researchers analyze a combination of interview data and archival data, based on the following three steps.

Firstly, we created vignettes based on the new value propositions from Gen-AI solution providers. The vignettes were based on the works of Miles & Huberman (1994), Pentland (1999), Seidman (2006), and Vaghefi et al. (2023). Please see Table 1 below for the analysis of companies in the interviews and Table 2 for the analysis of vignettes in the archival data; the category is shown in Appendix B. In addition to presenting the accounts of our interviewees, the vignettes were used to establish a chain of evidence between the empirical data and the concepts used to analyze them (Ologeanu-Taddei et al., 2023).

Secondly, we developed a data analysis framework (Figs. 1–5) based on the techniques of thematic analysis (Boyatzis, 1998) and data structure (Corley and Gioia, 2004) in order to establish a connection between the data and our theoretical concepts. Specifically, the research followed Gioia et al.'s (2013b) systematic, three-step approach to building a theoretical model with an inductive method with deductive elements. The first step involved identifying first-order categories in the data at the informant level, which emerged from consistent patterns in the data that constituted sufficient evidence for a characteristic practice. These practices were labeled based on each informant's language (Corbin and Strauss, 1990). In the second step, the first-order categories were grouped into second-order themes, which were then summarized as dimensions expressed in theoretical terms. This step generated a data structure that served as the foundation for the findings about Gen-AI's effects on new value propositions in BMI. Finally, the constant comparison method was used to continue comparing and updating the coding construct and theoretical framework, including analyzing data across informants, companies, and data sources (Glaser and Strauss, 2017). Once the coding table is constructed, the relational map is built following Langley's process theory approach (1999). Specifically, the authors apply visual mapping to enhance the validity of the process mapping by deploying 32 cases' value propositions for the relationships between case groupings (Langley, 1999).

Thirdly, we developed a generic narrative by utilizing the vignettes and the data analysis scheme proposed by Dunford and Jones (2000), Pentland (1999), and Riessman and Quinney (2005). The narrative depicts the gradual development of the evolving ecosystem of Gen-AI and BMI. Narratives are accounts of initiatives and their subsequent outcomes that are based on the focal firms' value propositions proposed by the informants (Ologeanu-Taddei et al., 2023). They clarify the causal logic of the targeted phenomenon, considering the perspectives of various stakeholders (Ologeanu-Taddei et al., 2023). As a result, authors aim to build the relational map consisting of "logically interconnected sets of propositions" that are "abstract enough to allow for generalizations but close enough to observed data to be empirically validated" (Hassan and Lowry, 2015; Hassan et al., 2023).

To ensure the reliability of the results, the researchers independently coded the interview data. Differences in coding and labeling based on the first-order codes from each researcher were discussed until a consensus was reached. A similar procedure was followed for determining and labeling second-order themes and dimensions. The discussions aimed to ensure that the findings were not solely based on a single researcher's analysis and that the findings represented an agreement between two researchers. This method helped to understand how Gen-AI affects new value propositions in BMI. Since its core components are practices that have been demonstrated to be effective and its emergent dimensions are theoretically rooted, the resulting framework responds to Thomas and Tymon's (1982) call for theory that is beneficial for both practice and conceptualization. Subsequently, the findings were categorized into three related sub-sections focusing on Gen-AI's effects on new value propositions in BMI.

Table 1 provides a comprehensive view of primary data from interviewed firms, showcasing a diverse spectrum of organizations within the IT sector. These firms, labeled F1 through F32, are involved in various industries such as software development, IT services and consulting, computer hardware manufacturing, technology, information, and internet services. The firms are geographically diverse, with headquarters spanning across the United States, Germany, Israel, Japan, the United Kingdom, South Korea, France, and the United Arab Emirates, and vary in size as indicated by employee numbers. The interviewees hold various pivotal roles, ranging from technical specialists to senior executives, providing a rich, multi-dimensional perspective on Gen-AI's influence. The value proposition categories, denoted by 'C', serve as a key reference to understand the specific aspects of BMI impacted by Gen-AI, reflecting its broad and multifaceted influence in the IT industry.

Table 2 provides an archival data profile of firms for the paper. This data represents secondary research sources, labeled as 'A' for archival data, and includes a diverse range of companies involved in Research Services, Technology, Information and Internet, Software Development, E-Learning, and IT Consulting. The firms, indexed from A1 to A14, are predominantly based in the United States but also include entities from the United Kingdom, China, and India, demonstrating a global perspective on Gen-AI's impact. Employee numbers vary, indicating a mix of small, medium, and large organizations. The category types, denoted by 'C', reference specific aspects or applications of Gen-AI within these organizations, illustrating the multifaceted influence of Gen-AI across different sectors and functions in the IT industry. These categories help us understand the broad spectrum of Gen-AI's application and its role in shaping new value propositions within diverse business models.

This study employed several techniques to strengthen its validity, including examining rival explanations and triangulating data sources and methods (Yin, 2013). To enable analytic generalization, the study aimed to extract more abstract ideas and explanations from the case findings that could be applied to other similar situations while carefully linking these generalizations to existing research literature and conducting replications of the case study where possible (Yin, 2013). The study's reliability was enhanced by using a case study protocol, developing a case study database, and maintaining a clear chain of evidence, allowing the procedures to be repeated with similar results.

4. Findings

4.1. Finding 1: Gen-AI affects new value propositions in BMI through 5 approaches

In this section, we provide evidence supporting each theme constituting the five emergent theoretical dimensions of the model. Please see Appendix C, which illustrates the whole data structure. We divide the whole data structure into five figures illustrating each dimension's data structure. Figs. 1 to 5 present the data structure of each dimension, showing the first-order categories (left) that generated the second-order themes (center), which we distilled into the overarching dimensions (right).

4.1.1. Dimension 1: Knowledge querying based on cloud solutions

Knowledge querying based on cloud solutions dimension is shown in Fig. 1.

Traditionally, cloud providers offer cloud solutions such as infrastructure as a service (IaaS) and platform as a service (PaaS). Now, some offer the add-on knowledge querying service based on Gen-AI to meet customers' demands that leverage AI algorithms and models to extract insights, query data, and make informed decisions. Word embedding represents words as vectors to capture semantic relationships in Natural Language Processing (NLP), which is an essential and huge amount of work in training AI models. In an interview with F4 (as shown in Table 1), the company states that customers do not want to create and run

Table 1
Primary interview firm profiles.

Index	Firm/Project description	Industry	Headquarter location	Employee number	Interviewee type	Value proposition
F1	Provides an MLOps platform for data science teams.	Software Development	US	201–500	Sales engineer leader	C18
F2	Develops NLP models and solutions.	Software Development	Germany	11–50	Product manager	C3
F3	Creates software for technical computing, simulation, and modeling.	Software Development	US	5001–10,000	Senior application engineer	C18
F4	Offers a unified platform for data management, analytics, and machine learning.	Software Development	US	1001–5000	Product marketing manager	C1
F5	Provides geospatial data visualization and analysis tools.	Technology, Information and Internet	US	51–200	Account executive	C19
F6	GPUs are for gaming and professional markets, and SoCs are for mobile computing and automotive markets.	Computer Hardware Manufacturing	US	10,001+	Account Executive	C17
F7	Gen-AI-powered video editor for near-real-time editing	Software Development	Israel	11–50	Cofounder	C8
F8	Offers a range of cloud computing, analytics, and AI solutions.	IT Services and IT Consulting	US	10,001+	Data scientist	C10, C18, C1, C2
F9	Provides IT services, including cloud computing, data analytics, and cybersecurity.	IT Services and IT Consulting	Japan	10,001+	Director of AI unit	C1, C18
F10	Provides cloud computing infrastructure and services.	Technology, Information and Internet	UK	1001–5000	Director of sales	C20
F11	Integrated Hardware solutions for AI	Computer Hardware Manufacturing	UK	51–200	Sales engineer	C17
F11	Licenses, and SaaS for Customized GPT solutions	Software Development	US	201–500	Data scientist	C3
F12	Provides cloud security solutions for cloud platforms.	Computer security	US	10,001+	Developer associate	C21
F13	Develops tools for analyzing and visualizing global news and events data.	Computer graphics	US	11–50	Developer associate	C19
F14	Creates product descriptions for various products and services.	E-commerce	US	1–10	Developer associate	C13
F15	Develops an automation script for the large language models	Gen-AI service provider	US	51–200	Developer associate	C4
F16	Human-centric, coherent whole program synthesis—your junior developer	Gen-AI service provider	US	11–50	Developer associate	C6
F17	AI-powered chatbots, smart devices, and cognitive search.	IT Services and IT Consulting	South Korea	51–200	CEO	C1, C3
F18	A technology distributor that provides hardware, software, and services to businesses.	IT Services and IT Consulting	US	1001–5000	Business development manager	C17
F19	A platform for labeling and managing data for machine learning models.	Software Development	US	51–200	Account director	C20
F20	Develops analytics, data management, and business intelligence software.	Software Development	US	10,001+	Business development specialist	C18
F21	A research and development institute that focuses on emerging technologies, including AI.	Research	UAE	201–500	Computer science researcher	C2
F22	Provides NLP tools for data privacy and security.	Software Development	US	51–200	Marketing specialist	C21
F23	Offers an enterprise AI platform for building, deploying, and managing AI models.	Software Development	US	1001–5000	Sales Engineer	C18
F24	Develops AI-powered tools for image and video editing.	Technology, Information and Internet	US	1001–5000	AI service director	C11
F25	Provides video conferencing and communication software.	IT Services and IT Consulting	US	5001–10,000	Business development manager	C12
F26	Provides cloud-based communication and collaboration solutions.	IT Services and IT Consulting	US	1001–5000	Business development specialist	C12
F27	Provides conversational AI chatbots for customer service and other applications.	IT Services and IT Consulting	UK	11–50	Director	C12

(*continued on next page*)

Table 1 (*continued*).

Index	Firm/Project description	Industry	Headquarter location	Employee number	Interviewee type	Value proposition
F28	Provides AI-powered customer service solutions.	Software Development	France	501–1000	Business development specialist	C12
F29	Provides a platform for building and deploying voice assistants.	Software Development	UK	11–50	CEO, founder	C12
F30	Provides a platform for building and deploying cross-platform applications.	Financial Services	US	51–200	Sales Manager	C12
F31	Provides voice and video communication solutions for the healthcare industry.	Telecommunications	UK	51–200	Technology and Engineering Manager	C12
F32	Provides customer service and experience management software.	Software Development	US	501–1000	Strategic Client Manager	C12

Table 2
Archival firm data.

Index	Company name	Industry	Headquarter location	Employee number	Category type
A1	A research company that develops artificial general intelligence (AGI).	Research Services	US	201–500	C3, C6, C9, C13, C14, C15
A2	An independent research lab exploring new methods of thought and human imagination.	Research Services	US	11–50	C13
A3	A platform that turns your ideas into videos.	Technology, Information and Internet	US	2–10	C8
A4	Autonomous AI agent available for free	Software Development	UK	2–10	C4
A5	open source, without cost for academic or business purposes large language model	Software Development	US	10,001+	C2
A6	AI software to detect artificially generated text	Technology, Information and Media	US	11–50	C5
A7	Writing AI detector is highly accurate at distinguishing AI from human writing.	Software Development	US	501–1000	C5
A8	Code completions as developers type and turn natural language prompts into code suggestions	Software Development	US	501–1000	C6
A9	Learn foreign language speaking with AI	E-Learning Providers	US	51–200	C9
A10	Build mindmap with AI	Software Development	China	11–50	C16
A11	Build the computing infrastructure for training AI	Computer Hardware Manufacturing	US	10,001+	C17
A12	AI supported cloud meeting solution	IT Services and IT Consulting	US	5001–10,000	C21
A13	Use AI to assist your PPT	Software Development	India	11–50	C7
A14	Make PPT with AI assistance	Software Development	US	11–50	C7

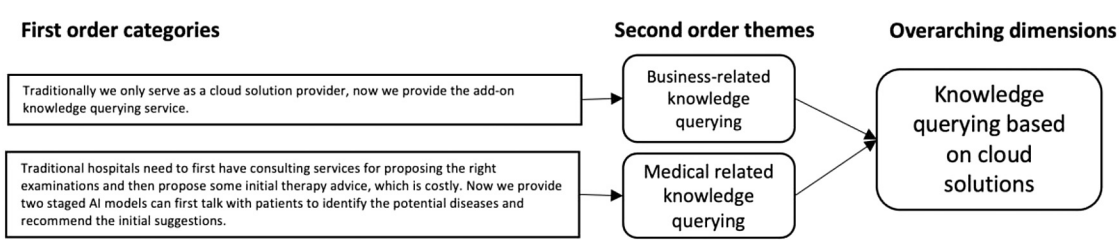

Fig. 1. Knowledge querying based on cloud solutions.

scripts to deploy the word embedding from the unstructured data in the transformation to the new services. So, the cloud provider builds user-friendly interfaces for customers to upload the data easily, utilize the existing data storage capabilities, and provide the add-on gen-AI capabilities.

Conventional hospitals typically rely on consulting services to recommend appropriate examinations and subsequently provide initial therapy advice, incurring significant costs. However, a two-stage Gen-AI model can now engage in initial conversations with patients, identify potential diseases, and offer preliminary suggestions. In an interview with F8 (as shown in Table 1), the leading AI solution company provides a question-answering computer system in healthcare for consulting and providing initial therapy advice through its Gen-AI computing capabilities. In this context, the system analyzes medical literature, patient records, and other relevant data to assist healthcare professionals in diagnosing illnesses and recommending suitable treatment options. The system can handle large volumes of medical information, extract meaningful insights, and offer evidence-based suggestions for patient care. This application aims to enhance the efficiency and accuracy of healthcare decision-making, ultimately improving patient outcomes and the overall quality of healthcare services.

4.1.2. Dimension 2: Foundation models

Foundation models are shown in Fig. 2.

Closed-source Gen-AI models that were traditionally proprietary are now being released as open-source, serving as foundation models

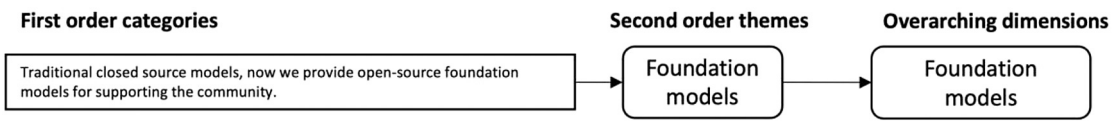

Fig. 2. Foundation models.

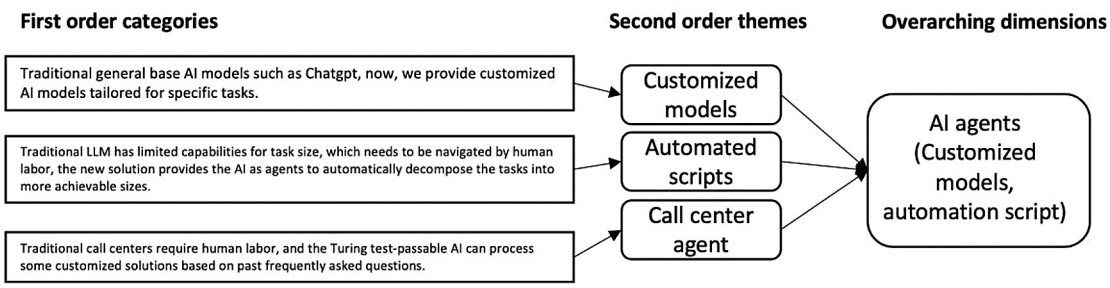

Fig. 3. AI agents (Customized models, automation script).

to support the broader community. The overall trend towards openness in Gen-AI models reflects a recognition of the benefits of shared knowledge and collaborative development. In an interview with F21 (as shown in Table 1), the research organization not only designs and develops its LLM but also makes it open-sourced to collaborate with other research institutions and companies on AI-driven projects.

4.1.3. Dimension 3: AI agents (Customized models, automation script)

AI agents (Customized models, automation script) are shown in Fig. 3.

Traditional general-purpose AI models like ChatGPT have evolved to offer customized AI models tailored for specific tasks. This shift allows for more targeted and efficient applications in various domains. It also provides control over style, ensures privacy, adaptability to tasks, faster deployment, cost-effectiveness, and flexibility for evolving requirements. In an interview with F11 (as shown in Table 1), the company provides licenses and software as a service (SaaS) for customized GPT solutions, which offer domain-specific expertise, enhanced performance, and improved accuracy.

The conventional LLM is constrained in handling larger tasks, necessitating manual intervention. The innovative solution introduces AI agents to break down tasks into more manageable sizes autonomously. Also, AI agents offer scalability, adaptability, and the potential for continuous improvement through machine learning, enhancing overall task performance and responsiveness. In an interview with F15 (as shown in Table 1), they efficiently break down extensive tasks into manageable parts, creating a framework that Gen-AI can adeptly handle and address. This collaborative approach enhances task management and performance.

Conventional call centers rely on human labor, while AI capable of passing the Turing test can generate customized solutions by processing information from frequently asked questions in the past. It also helps streamline customer interactions and support processes. In an interview with F28 (as shown in Table 1), Gen-AI solution brings flexible and customer-originated capabilities, including an efficient knowledge base and analytical services such as sentimental analysis compared with traditional call center solutions.

4.1.4. Dimension 4: Upstream industry chain infrastructure

Upstream industry chain infrastructure is shown in Fig. 4.

The conventional Graphics Processing Unit (GPU) was initially employed in the gaming industry. However, with evolving trends, its application has expanded to address diverse computational needs beyond gaming. With the growing demand for Gen-AI computing, there is a higher demand for GPUs and relevant hardware products. In an interview with F6 (as shown in Table 1), with the development of Gen-AI, the company gets more business with computer hardware, especially GPUs. They provide computing center solutions for many customers.

Traditional AI models combine tasks such as data processing, model selection, fine-tuning, testing, and evaluation across various platforms. Now, a unified machine learning operations (MLOps) solution simplifies this entire process. As the demand for Gen-AI computing rises, there is an increased need for MLOps. In an interview with F1 (refer to Table 1), the company experienced heightened business interest in the all-in-one solution for hosting machine learning algorithms with the development of Gen-AI.

Data labeling plays a crucial role in training AI models, aiding in creating accurate and effective algorithms by providing labeled examples for the model to learn from. Conventional data labeling solutions lack the flexibility to accommodate diverse domain-specific needs for training AI models. With Gen-AI, emerging data labeling services offer enhanced customization to meet specific labeling requirements. Furthermore, Gen-AI brings more demand for data labeling. In an interview with F23 (refer to Table 1), they provide specific data labeling services and accelerate custom model development. As Gen-AI evolves, the company experiences increased demand for data labeling services.

Conventional model training relies on proprietary data that is challenging to share because of privacy concerns, security risks, and the need to comply with regulations. In federated learning, models are trained collaboratively across multiple locations without centrally sharing raw data. This preserves privacy and security while enabling the aggregation of knowledge from diverse sources to improve the overall model. The collaborative nature of federated learning aligns with the principles of Gen-AI, promoting efficient and secure information exchange across a distributed network. Additionally, the rise of Gen-AI increases the need to share data. In an interview with F26 (refer to Table 1), Gen-AI provides synthetic data generation and sharing services. They stop the traceability of data sharing while maintaining statistical features. With the evolution of Gen-AI, the company witnessed a growing demand for data-sharing services.

4.1.5. Dimension 5: Content creation (Visual, text, audio, and analytical related)

Content creation (Visual, text, audio, and analytical related) is shown in Fig. 5.

Fig. 5 conceptualizes the transformative impact of Gen-AI on various domains of content creation. The categorization starts from specific applications and broadens to general themes, culminating in an overarching dimension. This analysis will explicate the second-order

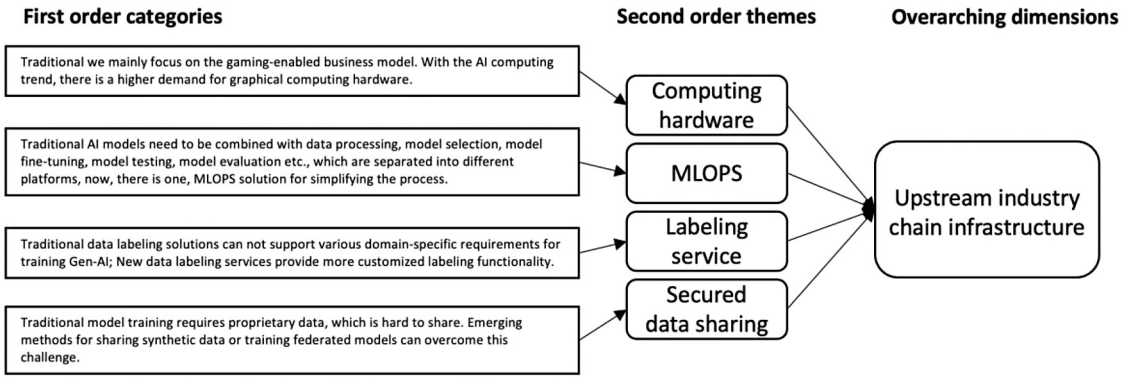

Fig. 4. Upstream industry chain infrastructure.

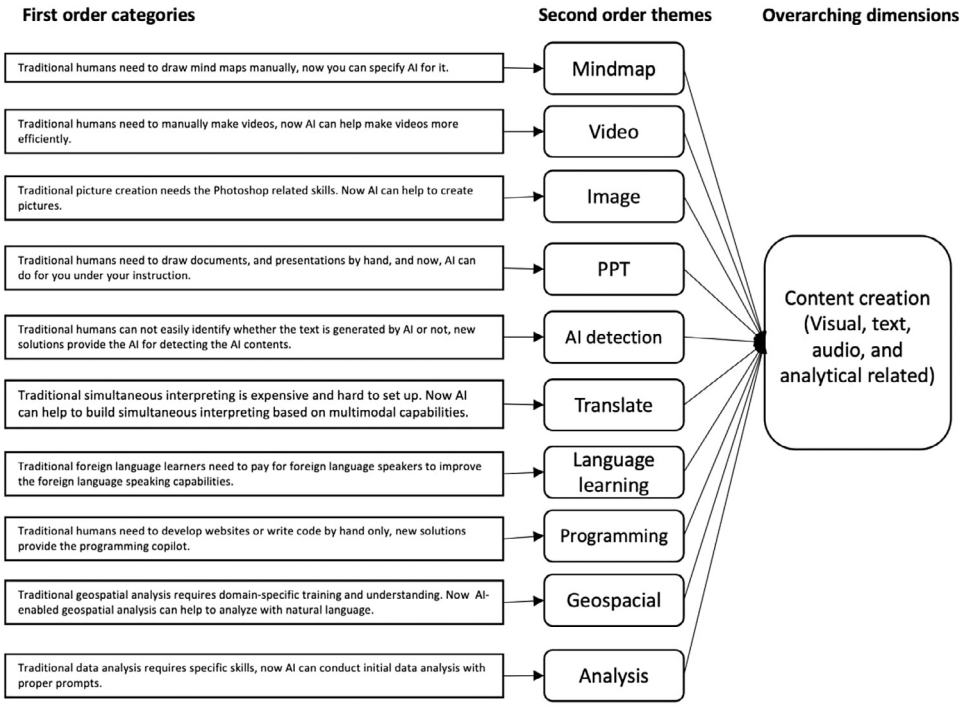

Fig. 5. Content creation (Visual, text, audio, and analytical related).

themes rooted in their corresponding first-order categories and culminate with examples, presumably from interviews, to illustrate the practical application of these AI advancements.

The theme of 'Mindmap' is drawn from the evolution of traditional manual mind mapping to AI-facilitated processes. Historically, individuals crafted mind maps manually, requiring significant time and effort. AI now enables users to direct the creation of these maps, thereby enhancing efficiency and effectiveness. The key concept here is the transition from manual to automated cognitive mapping, which allows for a more efficient visualization of ideas and their interrelations. For instance, project managers can now use AI to consolidate information from diverse sources into a coherent mind map, aiding in project planning and complex decision-making, as shown in A10 in Table 2.

In the realm of 'Video', the narrative shifts from manual video production to AI-assisted creation. Traditional video-making was a labor-intensive process that demanded a high level of skill. With AI, these processes are streamlined, allowing for more efficient editing and production. AI tools in video production can automate editing, suggest edits, and even create animations, significantly reducing the technical expertise required. An example from an interview with F7 in Table 1 highlights AI's ability to generate video content from brief text inputs, revolutionizing content creation in marketing and social media.

The 'Image' theme captures the shift from reliance on skilled professionals using complex software for picture creation to AI's ability to generate images based on textual descriptions. This key concept denotes a democratization of visual content production, enabling rapid prototyping and broadened creative exploration without the need for extensive training. The A2 in Table 2 shows how marketing teams leverage AI-generated images to quickly produce campaign visual content, bypassing the traditional need for graphic designers.

The 'PPT' theme represents the transition in document and presentation creation. Traditionally, crafting presentations was a hands-on task, requiring significant time investment. AI has transformed this process, allowing individuals to instruct AI to generate these materials. The central concept here is the automation of presentation creation, which can now incorporate complex data representation and design with minimal human input. Companies A13 and A14 in Table 2 use AI to auto-generate presentations filled with industry-specific data analytics, significantly enhancing productivity.

The 'AI detection' theme addresses the challenge of distinguishing AI-generated text from human-authored content. With the advent of AI, new solutions have been developed to detect AI-produced materials. This is a critical concept in maintaining the integrity and authenticity of digital content. For example, academic institutions utilize AI detection tools to identify submissions that are not the student's original work, which is shown in case A6, A7 in Table 2.

In the 'Translate' theme, the focus is on how AI has replaced expensive and complex simultaneous interpreting services with more accessible multimodal translation tools. The core concept here is the application of NLP for real-time translation, which has made such services more affordable and widely available. The application of AI provides real-time translation services across multiple languages with evidence from A1 in Table 2.

The 'Language learning' theme explores transitioning from traditional language learning methods, often involving costly tutors, to AI-driven, self-guided learning platforms. The central idea is that AI enables personalized language education, providing learners with adaptive resources and instant feedback. For example, A1 and A9 in Table 2 provide evidence of a user's experience with an AI language application that customizes lessons to their learning style, enhancing language acquisition.

Within the 'Programming' theme, the narrative shifts from manual coding to using AI as a programming assistant. Coding and website development once required extensive knowledge and experience, but AI now aids these tasks by suggesting code snippets and identifying errors. An interview with a developer associate in F16 in Table 1 highlights how AI assists in streamlining the development process, particularly in reducing time spent on debugging.

The 'Geospatial' theme encapsulates the evolution of geospatial analysis from a domain-specific expert task to an AI-enabled process accessible through natural language. This is significant in AI's capacity to interpret and analyze geographic data in user-friendly formats. An interview with an account executive in F5 in Table 1 reveals the utilization of AI to incorporate data from various sources for effective city planning and environmental assessments.

Finally, the 'Analysis' theme addresses how traditional data analysis, which necessitated specialized skills, can now be initiated by AI through user prompts. The underlying concept extends AI's capacity to include predictive modeling and trend analysis. The archive data of A1 in Table 2 reveals how they employ AI for initial data explorations to swiftly identify significant patterns and outliers.

The overarching dimension of 'Content creation' unifies these themes under the umbrella of AI-enhanced visual, textual, audio, and analytical content generation, indicating an integrative shift towards AI-assisted processes across diverse sectors. This synthesis suggests a broadening scope of AI's influence, from discrete tasks to comprehensive systems of innovation and creativity.

4.1.6. Finding 1 summary: Gen-AI's five value propositions for BMI

Finding 1 summary is shown in Fig. 6.

Fig. 6 delineates the multifaceted approaches that Gen-AI employs to craft new value propositions within the realm of BMI. It is articulated through a diagrammatic representation highlighting five distinct pathways, each marked by specific categories denoted by 'C', through which Gen-AI contributes to BMI.

The first pathway, 'AI agents (Customized models, Automation script)', identified as C3, C4, and C12, suggests that Gen-AI can offer tailored solutions by customizing models and automating scripts. It is also extensively discussed in the current IS literature (Han et al., 2023; Dennis et al., 2023; Chandra et al., 2022). This approach implies a move away from generic one-size-fits-all solutions towards more personalized and efficient automation strategies that can significantly enhance business processes and customer experiences.

The second approach, 'Content creation', comprising categories C5 through C9, and C11 to C16, C19, indicates that Gen-AI is pivotal in generating new content. In the literature, there are also related discussions on context generation with AI as decision support in the context of Tweet (Garvey et al., 2021). This approach includes textual and visual content and extends to the creation of multimedia and interactive materials. Such content creation facilitated by Gen-AI underpins novel product offerings and services, providing businesses with the tools to develop innovative ways to engage with their markets.

Thirdly, 'Knowledge querying based on cloud solutions', categorized as C1 and C10, highlights the role of Gen-AI in extracting and synthesizing information from vast data repositories in the cloud. In the literature, there is also a discussion about applying the knowledge graph for ecosystem intelligence (Basole et al., 2024). This capability underscores the potential of Gen-AI to revolutionize how businesses collect insights, make data-driven decisions, and tailor their offerings to meet the nuanced needs of their clientele.

The 'cloud solutions', denoted by C2, refers to the basic underlying structures and frameworks that Gen-AI can optimize or reinvent. In the literature, there has been discussion mainly on the enabling capabilities from foundation models in the domain-specific innovations such as multimodal cloud solutions (Fei et al., 2022). This foundational approach may involve rethinking the core algorithms or the structural aspects of how AI integrates into business operations, potentially redefining the baseline upon which further innovation can be built.

Lastly, the approach encapsulated in 'Upstream industry chain infrastructure', encompassing C17, C18, C20, and C21, addresses the impact of Gen-AI on the supply side of the business ecosystem. This approach echoes with the discussion of the importance of the cross-influences between hardware, software, and data (Vannuccini and Prytkova, 2023). It suggests that Gen-AI can transform not only the end products and services but also the upstream processes, such as sourcing, manufacturing, and logistics, thereby enabling a more coherent and streamlined business model.

These pathways converge to form new value propositions in BMI, suggesting that Gen-AI drives BMI. Gen-AI's interrelationship with these approaches shows its potential to transform business value creation, delivery, and capture. The figure illustrates how Gen-AI can be used to drive innovative business strategies and sustainable competitive advantages.

4.2. Finding 2: Gen-AI's effects on value propositions in BMI mapping for ecosystem and customization capabilities

Gen-AI's effects on BMI mapping for ecosystem and customization capabilities are shown in Fig. 7.

The diagram presents an analytical framework for understanding the influence of Gen-AI on BMI, focusing on the shift from upstream processes to downstream customization capabilities within the business ecosystem.

At the outset, positioned at the origin of the Gen-AI ecosystem's impact on BMI is 'Upstream industry chain infrastructure' (C17, C18, C20, C21). This denotes Gen-AI's foundational role in enhancing the upstream components of the value chain, including sourcing, manufacturing, and logistics. The emphasis here is on Gen-AI's capacity to streamline and optimize these core processes, which are critical to businesses' production and supply aspects.

Moving along the depicted curve, 'Foundation models' (C2) acts as a bridge between the upstream processes and more advanced applications. These models constitute the basic structures and algorithms that underpin Gen-AI, forming the architectural backbone that enables further innovation and customization.

In the Gen-AI ecosystem's continuum, 'Knowledge querying based on cloud solutions' (C1, C10) marks the initiation of a paradigm shift towards cloud-integrated data analytics. Gen-AI's proficiency in interrogating extensive cloud databases exemplifies the fusion of cloud computing with advanced data analytics, facilitating the derivation of nuanced insights. This modality, albeit customizable to a certain

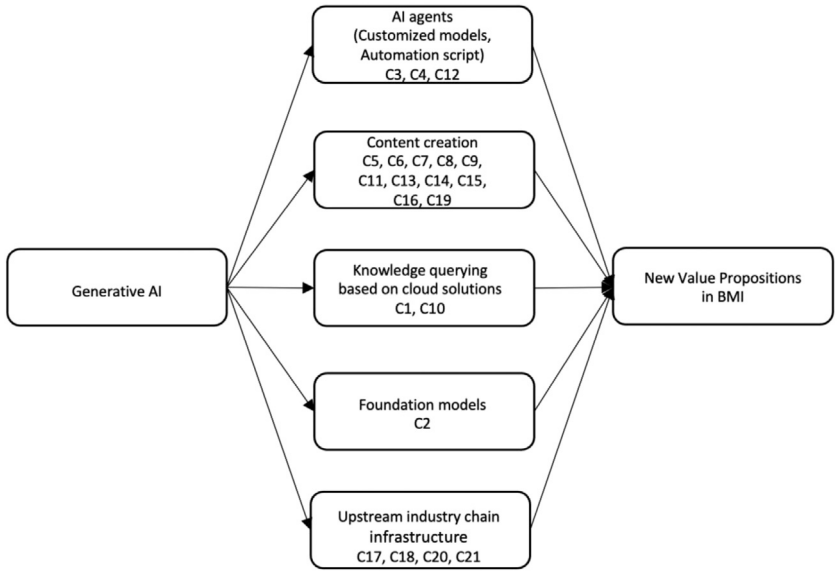

Fig. 6. Gen-AI's five approaches for new value propositions in BMI.

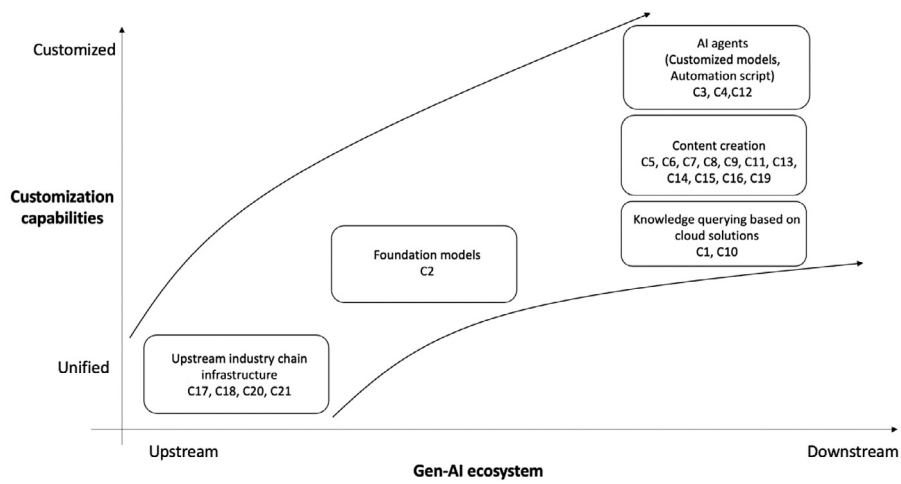

Fig. 7. Gen-AI's effects on BMI mapping for ecosystem and customization capabilities.

extent through data selection and query customization, presents a foundational layer of BMI. It enables enterprises to harness data-driven insights, thereby refining operational efficiencies and enhancing customer-centric strategies through an informed understanding of evolving market and consumer patterns.

Transitioning to 'Content creation' (C5–C9, C11, C13–C16, C19), Gen-AI ascends to a more dynamic and various field of innovation, where its capabilities extend to the generation of multifaceted digital content. This domain reflects a confluence of creativity and technology, with Gen-AI at the helm, producing content that is dynamic and reflective of consumer sentiments. The advent of Gen-AI in this sphere signals a pivot towards more bespoke content strategies that are pivotal for robust customer engagement and the propagation of tailored marketing narratives. Herein, the adaptability of Gen-AI becomes apparent, as it enables the tailoring of content to align with specific audience demographics and preferences, thereby intensifying the customization potential within BMI.

Culminating in 'AI agents' (C3, C4, C12), we observe the zenith of customization capability within the Gen-AI ecosystem. These specialized agents, tailored through bespoke models and automation scripts, are the pinnacle of personalization in BMI. Representing a symbiosis of Gen-AI's advanced potential with business-specific requisites, AI agents are instrumental in automating and refining complex processes, fostering the development of responsive services, and catalyzing customer interactions with unprecedented personalization. This high degree of customization encapsulates Gen-AI's transformative influence on BMI, endowing businesses with distinctive competencies to innovate and excel in a saturated marketplace.

The causal and temporal relationships could be further validated based on the chain of evidence exemplified by the cases (Mandviwalla, 2015). The upstream elements (C17, C18, C20, C21) serve as the foundational building blocks such as the computing infrastructure (GPUs and cloud solution providers) provider, temporally preceding and causally underpinning the emergence of the middle layer, which consists of foundation models (C2). It is because the foundation models nowadays have billions of parameters (Xu et al., 2022), and it would not even be possible without these advanced computing infrastructures. Moreover, hardware computing standards are more likely to be standard than diverse software-level application innovations. Therefore, the upstream industry chain infrastructure is in the unified upstream corner. Based on the computing infrastructure, the foundation models (exemplified by the C2) act as a crucial intermediary and the focal point of the discussion, bridging the upstream infrastructure with the downstream applications. Enabled by capabilities from these foundation models, the application-level solutions emerge, including AI

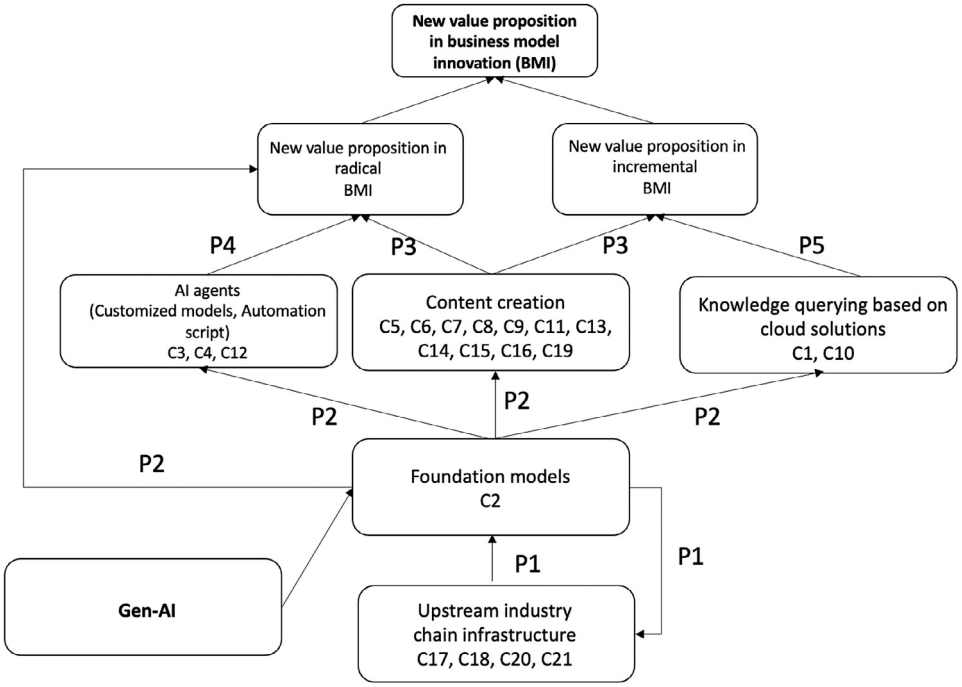

Fig. 8. Relational map of Gen-AI's effects on new value propositions in BMI.

agents (C3, C4, C12), content creation capabilities (C5, C6, C7, C8, C9, C11, C13, C14, C15, C16, C19), and knowledge querying based on cloud solutions (C1, C10). These solutions are more customized to the end user's tailored requirements, triggering the emerging innovation landscape (Yan et al., 2018) at the customized downstream section, depicted in Fig. 7.

In essence, Fig. 7 conceptualizes the Gen-AI ecosystem as a catalyst for innovation across the BMI spectrum. It enables a seamless flow from enhancing upstream processes to enabling downstream customization, thereby facilitating a comprehensive and nuanced approach to value creation and business model transformation.

5. Discussion

5.1. Theory development

Fig. 8 shows the relational map of Gen-AI's effects on BMI.

Fig. 8 depicts the dynamic interplay between Gen-AI and its multifaceted impacts on BMI. The visual model illustrates how Gen-AI exerts a cascading influence across different tiers of business model transformation — from incremental to radical innovations — while also emphasizing the iterative and reciprocal nature of these relationships (Ansari et al., 2016; Wan et al., 2015; Zhang et al., 2023).

At the foundation of this model is the 'Upstream industry chain infrastructure' (C17, C18, C20, C21), which, according to Proposition 1, is vital for advancing and operationalizing 'Foundation models' (C2). The diagram indicates that the capabilities of Gen-AI foundation models hinge on the support and maturity of upstream technological resources, including advanced computational provisions and infrastructural frameworks. In turn, these continuous advancements feed back into the Gen-AI ecosystem, driving ongoing cycles of improvement and innovation within business models, including the further iterative improvement of the upstream industry chain infrastructure.

Proposition 1. *Foundation models cannot achieve the existing intelligent capabilities without the upstream supply, such as higher graphical computing capabilities and modeling frameworks, including MLOps. The boost in foundation models has significantly increased the demand for the upstream supply chain.*

Ascending from this foundation, 'Foundation models' themselves are portrayed as pivotal in enabling a triad of downstream innovations: 'AI agents' (C3, C4, C12), 'Content creation' (C5–C9, C11, C13–C16, C19), and 'Knowledge querying based on cloud solutions' (C1, C10), as indicated by Proposition 2. This part of the model underscores the foundation models' role as a cornerstone upon which various forms of BMI are predicated, including the potential for these models to be disseminated as open-source resources, thus fostering communal growth and development within the AI sphere.

Proposition 2. *Foundation models could enable AI agents, content creation, and knowledge querying as a form of BMI. Additionally, foundation models can be open-sourced to support the AI community.*

In the realm of content generation, Fig. 8 situates 'Content creation' at an intermediate level of the model, reflecting Proposition 3's view of Gen-AI as an enabler of incremental BMI through the facilitation of diverse content formats. For example, Gen-AI has demonstrated enabling capabilities in the programming task (C6), document creation (C7), video creation (C8), language learning (C9), healthcare consulting (C10), picture creation (C13), simultaneous interpreting (C14), data analysis (C15), mind map creation (C16), and geospatial analysis (C19), which are shown in Table 3 together with the associated examples at the last column. Based on the primary interviews, the researchers found these enabling capabilities, although promising, need to be aligned with the human operators to provide the output that meets the required quality. In this case, these solutions could only be used as decision supports, as a form of operational level innovations, and have potential but yet limited influence on the organizational routines. Therefore, they are classified as incremental BMI (Velu, 2015; Friedman et al., 2008).

Additionally, including AI-generated content detectors implies a radical BMI element (C5, with the examples of A6 and A7). In this case, traditional methods cannot easily identify whether the text is generated by AI or not; the new proposed solution provides the AI for detecting the AI contents. Moreover, traditional video stock-selling businesses sell intellectual property (IP) for the video stocks. Now, they can provide AI-generated content and provide revenue back to the raw content creators based on the IP they created, which is a form

of radical innovation as well (C11, as exemplified by F24). They are examples of radical BMI, as they significantly change the organizational content creation routines and processes by providing new solutions and incentivization design (Velu, 2015; Friedman et al., 2008).

Proposition 3. *Content creators can use Gen-AI to provide new content in various formats (documents, videos, etc.) as an incremental BMI. Additionally, AI-generated content detectors are radical BMI. Protecting the IP raw content is essential, and this can be achieved through revenue-sharing options for the raw resource content creators.*

Transitioning to the upper level of innovation, 'AI agents' represent a form of radical BMI, as articulated in Proposition 4. The key rationale is that the advancement of Gen-AI has significantly extended the potential of AI agents' capabilities and boundaries. As suggested by Andrew Ng at the AI Ascent 2024, the agentic workflow and the reasoning capabilities could significantly improve the capabilities of LLM in coding challenges compared with a one-time response (zero-shot) of LLM.[6] These collaborative AI agents enable a new domain of innovations by utilizing the innovative solution of the single Gen-AI for achieving the synergy effects (Wu et al., 2023). The model emphasizes the transformative nature of these agents, positing that custom AI solutions and autonomous frameworks could surpass human capabilities, heralding a potential paradigm shift in business operational models. Therefore, AI agents could enable new value propositions such as radical BMI.

Proposition 4. *Customized AI and autonomous AI frameworks that provide AI agents that could be equal to or better than human individuals are forms of radical BMI, as it has not existed before and has a huge potential for changing how business works.*

Finally, per Proposition 5, 'Knowledge querying based on cloud solutions' is depicted as an extension of incremental BMI, reflecting the role of cloud service providers in offering Gen-AI-powered knowledge querying services. It is because cases C1 and C10 are existing cloud solution providers that provide existing knowledge searching and querying services. With the evolving capabilities of Gen-AI, cloud solution providers such as C1 and C10 could provide additional add-on services for building in-depth knowledge querying based on the customer's proprietary data. This form of add-on service innovation could be classified as a new value proposition in forming the incremental BMI. This proposition suggests an evolutionary rather than revolutionary impact on BMI, enhancing analytical capabilities and business intelligence through cloud-based AI services.

Proposition 5. *Cloud solution providers provide Gen-AI as knowledge querying services, as incremental BMI.*

In essence, Fig. 8 encapsulates a comprehensive relational map that illustrates Gen-AI's escalating impact from foundational technologies to highly specialized applications within BMI. It visually represents how each proposition feeds into the next, culminating in a layered portrayal of Gen-AI's potential to drive both incremental and radical innovation within the business landscape.

5.2. Contribution to literature

The traditional view of Gen-AI's value propositions is mainly related to the content creation (Epstein et al., 2023; Wessel et al., 2023; Kanbach et al., 2023), such as generating various forms of documents. This research has confirmed these new value propositions. Additionally, it has extended with four new value propositions from Gen-AI in BMI: knowledge querying-based cloud solutions, content creation, AI agents, foundation models, and upstream industry chain infrastructure, as shown in finding 1. This discussion has only occurred in the computer science or AI literature (Baidoo-Anu and Ansah, 2023; Ooi et al., 2023); we are extending this discussion to the field of management and business model-related literature.

Additionally, existing discussion in the management literature has been extensively focused in the higher education field (Michel-Villarreal et al., 2023; Noy and Zhang, 2023), with some survey data from the operations field (Wamba et al., 2023). This research has provided empirical evidence based on semi-structured interviews in the information technology industry in illustrating Gen-AI new value propositions in BMI, specifically in the IT industry, which extends the contextual boundaries of the discussion.

Furthermore, existing literature has only focused on some potential new mechanisms of Gen-AI for BMI (Kanbach et al., 2023; Gursoy et al., 2023). However, the mechanisms' inter-related dynamic effects have not been fully evaluated. Findings 2 and 3 in this paper provide the quadrant view and process map for Gen-AI new value propositions in BMI, which consisted of radical BMI and incremental BMI. It goes beyond the five theoretical dimensions to the inter-related dynamic relationship, represented by the propositions as our contribution to literature.

5.3. Theoretical implications

The main theoretical contribution of this study is to provide early empirical evidence on how Gen-AI creates new value propositions in BMI. The findings expand Velu's (2018) theory of value propositions in BMI to Gen-AI context. The study identifies five possible ways Gen-AI could create new value propositions: improving upstream industry chain infrastructure, developing foundation models, creating AI agents, enabling content creation, and offering knowledge querying through cloud solutions. The research also suggests that these Gen-AI-enabled value propositions could lead to both incremental and radical BMI, which is a new idea that builds on Velu's (2015) work on different types of BMI. By showing how Gen-AI's value propositions enable different kinds of BMI, this study offers a preliminary exploration of how Gen-AI and BMI are related. Additionally, this research has extended the discussion of Gen-AI from the domain-specific contexts, into the upstream solution providers' perspectives as the down-stream solutions providers are dependent on the solutions from upstream in the context of Gen-AI.

Moreover, examining these findings through a socio-technical lens further enriches our theoretical understanding. A socio-technical perspective foregrounds the interplay between Gen-AI's technical design (e.g., foundation models, AI agents, and data infrastructures) and the social context in which it is embedded (e.g., organizational behaviors, employee skill sets, ethical norms, and regulatory frameworks). From this viewpoint, the efficacy of Gen-AI in driving BMI is not simply determined by its computational power or the novelty of the value propositions it enables, but rather by the alignment between technical capabilities and the social structures that shape technology adoption and use. As organizations introduce Gen-AI-based services — such as cloud-based knowledge querying or autonomous AI agents — they must consider human factors like trust, workforce readiness, and potential changes in job roles. Likewise, navigating ethical and privacy implications requires clear organizational policies and stakeholder collaboration. These socio-technical considerations reinforce that BMI in Gen-AI contexts goes beyond purely technological innovations; it necessitates co-design with social and organizational processes to ensure sustainable and responsible deployment.

In summary, the five ways Gen-AI could create value found in this research provide a starting point for thinking about how Gen-AI might affect value propositions in BMI (Fig. 8). Each approach is a possible path Gen-AI could take to change how value is created, from improving upstream supply chain processes to enabling highly customized AI solutions. By mapping out these potential paths, the study contributes to an early theory of how Gen-AI could shape BMI.

[6] https://www.sequoiacap.com/article/ai-ascent-2024/

5.4. Contribution to practice

The practical contribution is a critical element for qualitative theory building (Geletkanycz and Tepper, 2012). This research provides a practical framework for business stakeholders to understand how Gen-AI enables new value propositions. The framework offers an ecosystem view of the types of business solutions that can be provided (Fig. 6), their mapping within the Gen-AI ecosystem (Fig. 7), and their dynamic interplay towards incremental and radical innovations as a relational map (Fig. 8). Specifically, Figs. 6 and 7 could trigger new BMI opportunities. For example, for IT solution providers, our study provides a roadmap for leveraging Gen-AI for BMI. Companies can start by experimenting with foundation models and knowledge querying to enhance existing products and processes. On the other hand, For domain-specific companies (potential customers of the IT solution), such as manufacturing and service providers, this research provides an overview of the existing capabilities enabled in the IT industry for them to better upgrade and integrate the potential solutions for domain-specific new value propositions. Over time, they can invest in developing customized AI solutions and content generation capabilities to unlock new revenue streams and customer segments. This insight can help businesses make informed decisions about adopting and integrating Gen-AI technologies to drive BMI.

This study also has policy implications for Gen-AI governance and regulation in business. As Gen-AI adoption accelerates and disrupts business models across industries, policymakers must develop legal and regulatory frameworks to ensure responsible development and deployment. This study provides the relational map for Gen-AI's new value propositions in enabling the BMI (Fig. 8), which could better position the policymakers in issuing the targeted regulations. It will further help to achieve the healthy and sustainable growth of Gen-AI business field.

6. Conclusions

6.1. Conclusion of the study

This study aimed to investigate Gen-AI's effects on new value propositions in BMI in the IT industry. The goal was to enhance our understanding of how value propositions may be achieved using Gen-AI. This study specifically examined the mechanisms and process dynamics for new value propositions in the Gen-AI field, leading to new value propositions in both radical and incremental BMI. The study's primary findings can be briefly described and concluded as follows: (1) Gen-AI affects new value propositions in BMI through five approaches: the upstream industry chain infrastructure, foundation models, AI agents, content creation, and knowledge querying based on cloud solutions; (2) Gen-AI's effects on value propositions in BMI can be represented in an evolutionary mapping from the upstream unified customization capabilities as industrial chain infrastructure, to the foundation models as the focal point, to the downstream higher customized solutions with knowledge querying, content creation and AI agents. (3) The relational map of Gen-AI's effects on BMI comprehensively describes these findings.

- Foundation models cannot achieve the existing intelligent capabilities without the upper stream supply, such as higher graphical computing capabilities and modeling frameworks, including MLOps. The boost of foundation models has significantly increased the demand for the upstream supply chain.
- Foundation models could enable AI agents, content creation, and knowledge querying as a form of BMI. Additionally, foundation models can be open-sourced as a form of BMI to support the AI community.
- Content creators can use Gen-AI to provide new content in various formats (documents, videos, etc.) as an incremental BMI. Additionally, AI-generated content detectors are radical BMI. Protecting the IP raw content is essential, and this can be achieved through revenue-sharing options for the raw resource content creators.
- Customized AI and autonomous AI frameworks provide AI agents that could be equal to or better than human individuals are forms of radical BMI, as it has not existed before and has a huge potential for changing the way business works.
- Cloud solution providers provide Gen-AI as knowledge querying services, as incremental BMI.

6.2. Limitations

One limitation of the study is that qualitative research methods were used to collect and process data. Qualitative methods can provide a deep understanding of complex issues and help explore new and under-researched areas, but they may be limited by researcher bias in data interpretation and limited generalizability compared to quantitative methods.

Furthermore, the study collected empirical data primarily from the IT industry, which is increasingly developing Gen-AI solutions to empower various potential businesses. Thus, when generalizing the findings, industry-specific attributes must be acknowledged. The IT industry's focus on Gen-AI and its unique characteristics may limit the findings' applicability to other industries without further validation. To improve generalizability, researchers could explore Gen-AI's impact on BMI in more industries.

The study mostly used data from UK, EU, and US companies. The sample's geographic concentration may limit the findings' global representativeness, even though these regions are leading Gen-AI development. Similar studies in other regions, such as Asia, the Middle East, and Latin America, where Gen-AI is also rapidly advancing, could help better understand Gen-AI's impact on BMI across cultural and economic contexts.

6.3. Future research

Future researchers could provide a comparative multi-case study with case firms not only from the UK, EU, and the US but also cases from Asia or other regions (Rantala et al., 2023). In future multi-case studies, we will provide comparative dimensions to disentangle unique new value propositions based on carefully designed within- and cross-case analysis.

Additionally, further research is to be conducted with respect to Gen-AI's effects on the value creation, value capture, and value network in the context of BMI. Quantitative methodologies could also be considered for a more robust evaluation of the propositions. Specifically, as the prevalence of Gen-AI increases, it is worth considering the potential for value propositions by utilizing the signaling effects of Gen-AI announcement through an event study.

CRediT authorship contribution statement

Dequn Teng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chen Ye:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Veronica Martinez:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Informed consent

Verbal consent has been obtained from the interviewees.

Human and animal participants

This research is deemed extremely low risk of harm to human participants because of reasonably handling of personal data/anonymity.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly and ChatGPT in order to check grammar and polish the manuscript. After using these tools/services, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Appendix A. Questions

- Background and introduction
  - What does your company do?
  - Where is your company's headquarter?
  - What is the range of employees the company has?
  - What is your role in the company?
- Gen-AI adoption
  - Gen-AI is exemplified by ChatGPT, Midjourney, etc., for providing AI assistance as a service.
  - Have you seen Gen-AI adoption in the firm?

Table 3
Category referencing table.

Category type	Value proposition	Examples
C1	Traditionally only as a cloud solution provider, now provide the add-on knowledge querying service	F4, F8, F12, F9, F17
C2	Traditional closed-sourced models now provide open-sourced foundation models for supporting the community	F21, F8, (A5)
C3	Traditional general-based AI models, such as ChatGPT, now provide customized AI models tailored for specific tasks.	F11, F2, F17 (A1)
C4	Traditional LLM has limited capabilities for task size, which needs to be navigated by human labor; the new solution provides the AI as agents to automatically decompose the tasks into more achievable sizes.	F15, (A4)
C5	Traditional human cannot easily identify whether the text is generated by AI or not, new solution provide the AI for detecting the AI contents.	(A6, A7)
C6	Traditional humans need to develop websites or write code by hand only; new solutions provide the programming copilot	(A8, A1), F16
C7	Traditional humans need to draw documents and presentations by hand, and now, AI can do it for you under your instruction.	(A13, A14)
C8	Traditional humans need to manually make videos, now AI can help make videos more efficiently	F7 (A3)
C9	Traditional foreign language learners need to pay for foreign language speakers to improve their foreign language speaking capabilities	(A1, A9)
C10	Traditional hospitals need to first have consulting services to propose the right examinations and then propose some initial therapy advice, which is costly. Now, two staged AI models can first talk with patients to identify potential diseases and recommend initial suggestions.	F8
C11	Traditional video stock-selling businesses sell the IP for the video stocks. Now, they can provide the AI-generated content and provide revenue share back to the raw content creators	F24
C12	Traditional call centers require human labor, and the Turing test-passable AI can process some customized solutions based on the past frequently asked questions	F25, F26, F27, F28, F29, F30, F31, F32
C13	Traditional picture creation needs the PS-related skills. Now AI can help to create pictures	(A2, A1), F14
C14	Traditional simultaneous interpreting is expensive and hard to set up. Now, AI can help build simultaneous interpreting based on multimodal capabilities.	(A1)
C15	Traditional data analysis requires specific skills. Now, AI can conduct initial data analysis with proper prompts	(A1)
C16	Traditional human need to draw mind map manually, now you can specify AI for it.	(A10)
C17	Traditional business development mainly focuses on the gaming-enabled cases. With the AI computing trend, there is a higher demand for graphical computing cards	F6, F11, F18, (A11)
C18	Traditional AI models need to be combined with data processing, model selection, model fine-tuning, model testing, model evaluation, etc., which are separated into different platforms. Now, there is one MLOps solution to simplify the process.	F1, F3, F8, F9, F20, F23
C19	Traditional geospatial analysis requires domain-specific training and understanding. Now, AI-enabled geospatial analysis can help natural language analysis with open-source models.	F5, F13
C20	Traditional data labeling solutions cannot support various domain-specific requirements for training Gen-AI; New data labeling services provide more customized labeling functionality	F10, F19
C21	Traditional model training applies proprietary data, which is hard to share. Emerging methods for sharing synthetic data or training federated models can overcome this challenge.	F22, F12, (A12)

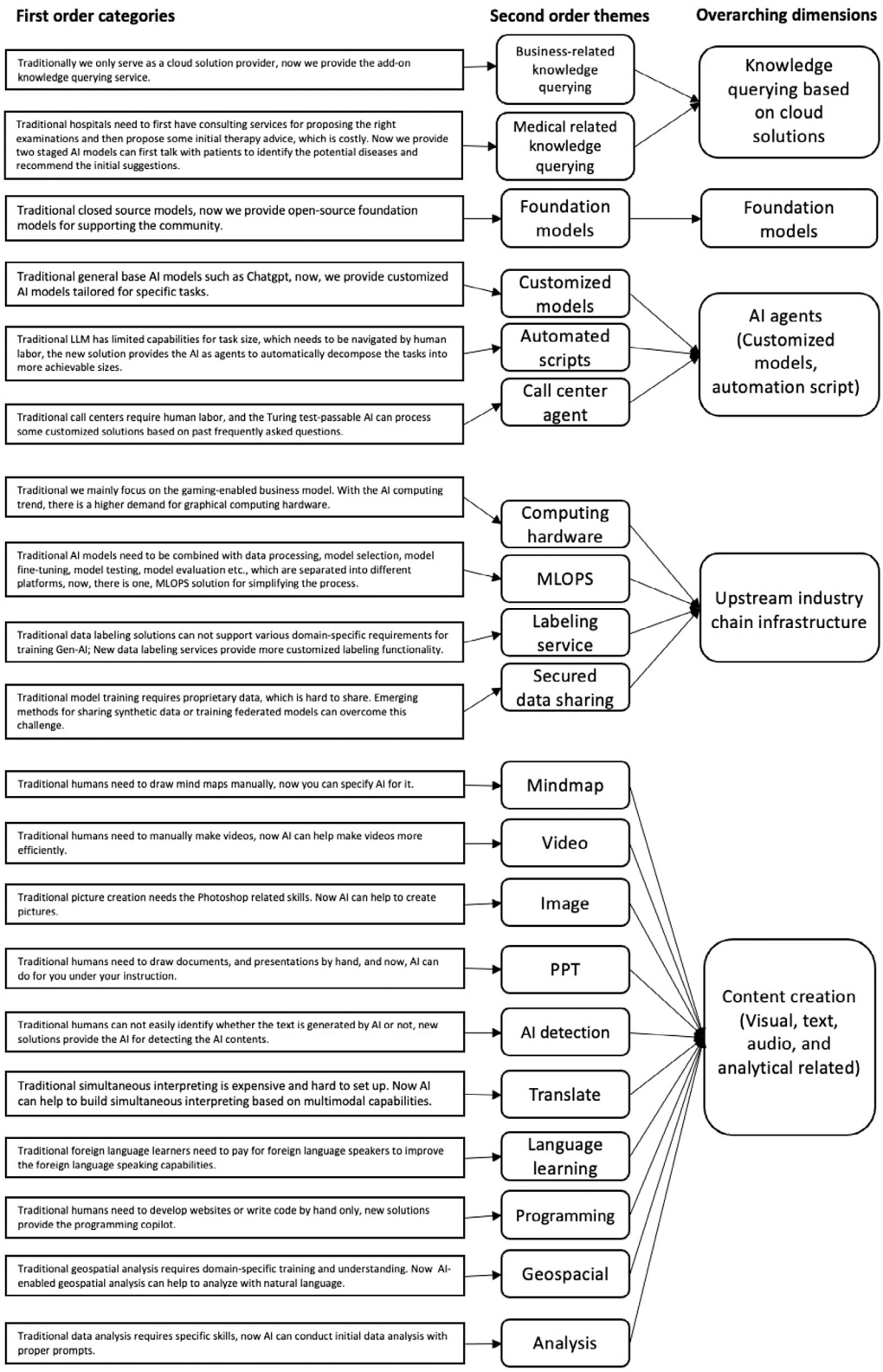

Fig. 9. Aggregated coding table.

- If so, by how?
- If not, are you planning to?
  - Could you elaborate on the example?

- Value propositions in business model innovation

Table 4
Data sources.

Data types	Collection details and dates	Page
Primary Data Semi-structured Interview	32 interviews with in total 18.2 h of transcripts from Mar 2023 to Dec 2023	Transcription (English) from recordings
Government reports	Whitehouse's reports on Gen-AI	2 documents, 118 pages
	European report on Gen-AI	1 document, 20 pages
	UK government's reports on Gen-AI	3 documents, 121 pages
Consulting reports	Deloitt's reports on Gen-AI	4 documents, 68 pages
	IBM's reports on Gen-AI	5 documents, 73 pages
	Mckinsey's reports on Gen-AI	2 documents, 92 pages
	EY's reports on Gen-AI	8 documents, 163 pages
	PwC's reports on Gen-AI	6 documents, 41 pages
	Bain's reports on Gen-AI	6 documents, 237 pages
	BCG's reports on Gen-AI	5 documents, 67 pages
	KPMG's reports on Gen-AI	5 documents, 63 pages
	Accenture's reports on Gen-AI	2 documents, 71 pages
	Oliverwyman's reports on Gen-AI	8 documents, 190 pages
	Euromonitor's reports on Gen-AI	2 documents, 14 pages
Market research	Gartner's reports on Gen-AI	2 documents, 15 pages
	Statista's reports on Gen-AI	2 documents, 13 pages
Corporates' reports	Capgemini's reports on Gen-AI	3 documents, 228 pages
	SAP's reports on Gen-AI	3 documents, 27 pages
NGOs' reports	The World Economic Forum's report on Gen-AI	1 document, 24 pages
	CEPR's reports on Gen-AI	4 documents, 39 pages
	Economist's reports on Gen-AI	6 documents, 30 pages
Media	Forbes' reports on Gen-AI	8 documents, 52 pages
	The Wall Street Journal's reports on Gen-AI	5 documents, 25 pages
Books	Books about Gen-AI	2 books, 416 pages
Videos	Videos about Gen-AI	12 Youtube videos

- Value propositions in the context of business models are defined as a company's products or services delivered to the customers (product/service you offer).
- Have you changed the value propositions compared with prior practices as a result of Gen-AI adoption as an intervention?
  - If you, by how?
  - If else, have you heard some evidence in the industrial sector you operate in?
- Could you elaborate on the example?

Appendix B. Category referencing table

Table 3 shows the category referencing table.

Appendix C. Aggregated coding table

The aggregated coding table is shown in Fig. 9.

Appendix D. Data sources

The data sources are shown in Table 4.

Data availability

The authors do not have permission to share data.

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